5 November 1969
SUPERSEDING
MIL-T-38491 (USAF)
19 July 1967

MILITARY SPECIFICATION

TEST SET, GUIDED MISSILE LAUNCHER A/E24T-81

1. SCOPE

1.1 This specification covers a launcher test set designated as Guided Missile Launcher Test Set A/E24T-81 hereinafter referred to as "Test Set" (for operational use reference Section 6 of this specification).

*2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-591	Boxes, Fiberboard, Wood-Cleated
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Box, Fiberboard
PPP-B-640	Boxes, Fiberboard, Corrugated, Triple-Wall

Military

MIL-P-116	Preservation, Methods of
MIL-M-4552	Multimeter AN/PSM-6B
MIL-L-10547	Liners, Case and Sheet, Overwrap; Water-Vaporproof or Waterproof Flexible

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Military

MIL-N-18307	Nomenclature and Nameplates for Aeronautical, Electronic and Associated Equipment
MIL-T-21200	Test Equipment for Use with Electronic and Fire Control Systems, General Specification for
MIL-I-45208	Inspection System Requirements

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-480	Configuration Control - Engineering Changes, Deviations and Waivers
MIL-STD-810	Environmental Tests Methods for Aerospace and Ground Equipment
MIL-STD-831	Test Reports, Preparation of

OTHER DOCUMENTS

AFSCM/AFLCM 310-1	Management of Contractor Data and Reports
ANA Bulletin 438	Age Controls of Age-Sensitive Elastomeric Items

DRAWINGS

Air Force (AF) Drawing Number (No.) 65J17174

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

*3.1 Qualification. First Article tests of three Test Sets shall be made in accordance with the provisions contained herein.

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3.2 General specification. The requirements of MIL-T-21200 for Class 2 equipment shall apply as requirements of this specification with the exceptions or additions called out herein.

3.2.1 Nameplate. Approval of the drawing for the nameplate shall be by the Engineering Division of the Procuring Activity. The format shall essentially conform to Figure 3 of MIL-N-18307.

3.3 Service conditions. The equipment shall operate with no degradation of performance under any combination of service conditions specified in MIL-STD-810, Table 1, Column E. The design criteria shall be those applicable to Ground Equipment and world-wide operation.

3.4 Design.

3.4.1 Configuration. The dimensions of the Test Set and cables shall be in accordance with AF Drawing No. 65J17174.

3.4.2 Design reference. AF Drawing No. 65J17174 shall be used as a reference and guide for design of the Test Set.

3.4.3 Components. The Test Set shall comprise one complete unit which shall include the chassis with control panel, cable assembly and attaching provisions for an AN/PSM-6 Meter. Selection of components and design criteria shall be in accordance with the provisions of MIL-T-21200.

3.4.4 Attaching provisions. Two catches shall be provided on the chassis bracket for physically attaching an AN/PSM-6 Meter. Two leads with connectors shall be provided for electrical connection to the AN/PSM-6 Meter.

3.4.5 Electrical requirements. The Test Set shall be designed for use with voltages up to 115 Volts Alternating Current (VAC), and shall be capable of providing the switching necessary to verify proper operation of the launcher firing circuits when used with an AN/PSM-6 Meter. Reference AF Drawing 65J17174.

3.4.6 Control panel. The Control Panel shall be marked as shown on AF Drawing No. 65J17174, and in accordance with MIL-T-21200.

*3.4.7 Sealing gaskets. A gasket shall be provided between the chassis and the control panel and between the chassis and the bottom panel. The gasket shall be fungus resistant.

*3.5 Engineering changes. Engineering changes to the requirements of this specification or to design and manufacturing drawings and contractor's control specifications after approval of preproduction Test Sets shall conform to the requirements of MIL-STD-480.

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3.6 Contractor furnished equipment. The supplier shall furnish the test device described in 4.4.3 to be used for operational tests on completed Test Sets.

3.7 Government loaned property. When specified in the contract, the Government will loan the following to the contractor upon his request.

ITEM NO.	STOCK NO.	PART NO.	ITEM DESCRIPTION	QUANTITY
1	6625-643-1686 or Equal	MIL-M-4552 Type AN/PSM-6	Multimeter w/ Accessories	1

4. QUALITY ASSURANCE PROVISIONS

*4.1 Responsibility for inspection. Unless otherwise specified in the contract, or purchase order, the supplier is responsible for the performance of all inspection and test requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections or tests set forth in this specification where such inspections or tests are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test procedures. Detailed Test Procedures shall be prepared by the contractor for each classification of tests and inspections. Five copies of each procedure shall be submitted to the Procuring Activity in accordance with AFSCM/AFLCM 310-1 as specified in Section 6 of this specification for engineering review and approval after award of contract and prior to start of testing.

*4.1.2 Test reports. The contractor shall submit five copies of First Article Test Reports with the First Articles to the Procuring Activity in accordance with AFSCM/AFLCM 310-1 as specified in Section 6 of this specification for engineering review and approval after completion of First Article Testing as specified in the contract. Test Reports shall be prepared in accordance with MIL-STD-831. A complete set of contractor's production, process and control drawings, schematic, and contractor specifications for non-standard parts shall be submitted with the First Articles and test reports submitted for engineering approval.

*4.2 Classification of inspections. The inspection and testing of the Test Sets shall be classified as follows:

a. First Article Tests (4.3)

b. Quality Conformance Tests (4.4)

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*4.3 First article testing.

*4.3.1 First article equipments. First Article Tests shall be accomplished on three Test Sets representative of the Production Test Sets. The Test Sets shall be identified with Nameplate and complete identification data in same manner as production Test Sets, including serial number identification.

*4.3.2 First article tests. The First Article Tests are intended to provide assurance that the design and operation of the Test Set will meet the requirements. Successful completion of the test program does not relieve the contractor of the responsibility to furnish equipment that meets the performance requirements of this specification. The supplier shall furnish test reports in accordance with AFSCM/AFLCM 310-1 as specified in Section 6 of this specification showing quantitative results for all tests, and shall be signed by an authorized representative of the contractor or laboratory, as applicable. An operational test shall be performed before and after each Environmental Test. First Article Tests shall consist of Quality Conformance Tests of 4.4 and the following Environmental Tests of MIL-STD-810, Table I, Column E. Where this standard requires operation of the test item, the Operational Tests of 4.4.3 and Insulation Tests of 4.4.4 shall be performed.

a. Low Temperature, Method 502, Procedure I. Lower storage limit is -62° Centigrade (C) or -80° Fahrenheit (F) for 72 hours. Lower operational limit is -40°C (-40°F).

b. High Temperature, Method 501, Procedure I. Higher storage limit and operational limit is 71°C (160°F).

c. Altitude, Method 500, Procedure I.

d. Dust, Method 510, Procedure I. Test specimen will not be operated during test. Time interval between step 2 and step 3 will be 0 to 16 hours at the contractor's option.

e. Shock, Method 516, Procedure I. Test Set shall be installed in transit case or package. Apply shock per Figure 516-1, peak value 20 g's, duration 11 milliseconds. Operation of Test Set during test is not required.

f. Vibration, Method 514, Procedures X (curve AB) and XI (see Table 514-1, Row h). Test from 2-500 Hertz (Hz).

g. Rain, Method 506, Procedure I. Operation of Test Set during test is not required.

h. Humidity, Method 507, Procedure I.

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i. Fungus, Method 508, Procedure I. Test Set shall be operated after fungus test.

j. Salt Fog, Method 509, Procedure I.

4.4 Quality conformance tests. Quality Conformance Tests are those tests performed during production to verify the integrity of equipment, quality of workmanship and operational performance for compliance with the requirements of this specification. The contractor shall furnish all equipment except that which is authorized for bailment herein. All inspection and testing shall be under the supervision of the Government Representative. Quality Conformance Tests shall consist of the following:

a. Examination of Product

b. Continuity Test

c. Operational Test

d. Insulation Test

e. Environmental Sampling Test

4.4.1 Examination of product. Each Test Set shall be examined to determine conformance to this specification with respect to materials, workmanship, connection to AN/PSM-6 Meter, and general requirements.

4.4.2 Continuity test. A Continuity Test shall be conducted on each Test Set to demonstrate compliance with the continuity requirements of AF Drawing No. 65J17174.

*4.4.3 Operational tests. Each Test Set shall be given an operational test to verify proper operation. The Test Set shall be attached to a Government furnished AN/PSM-6 Multimeter or equivalent. The connector P701 and P700 shall be connected to a suitable contractor-furnished device which provides voltages of 115 ± 6 VAC, 400 Hz and 28 ± 1.5 Volts Direct Current (VDC) in the following manner:

a. 115 VAC between P701-C, B and P701-A

b. +28 VDC to P701-H, Z, L, P, E, and P700-A

c. -DC to P701-A and shell of P700

4.4.3.1 The rotary switch shall be rotated through its 10 positions and proper readouts verified on the AN/PSM-6 Meter or equivalent (reference AF Drawing No. 65J17174).

4.4.3.2 With audio test switch depressed, the voltage measured between P701-U to P701-A shall be 4.0 ± .5 VAC 400 Hz.

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4.4.3.3 With the rotary switch rotated to position 10 the voltage measured between P701-V, W and P700-X, Y shall be 2.4 VDC. (This voltage was adjusted in switch position 9.)

*4.4.4 Insulation tests. With the Test Set disconnected from the AN/PSM-6, the rotary switch in position #9 and the interlock switch in "off" position (except as noted below), a megger with high potential of 500 volts minimum shall be applied only to the shell of connectors P700, P701 and Pins P700-A; P701-H, R, Z, L, P, E, B, W (reference AF Drawing No. 65J17174) as follows:

a. Between each connector pin or shell mentioned above and every other connector pin or shell mentioned above.

b. Between each connector pin or shell mentioned above and the Test Set Case except between shell P700 and Test Set Case when interlock switch shall be in the "ON" position.

c. The voltage shall not be applied to Pins P701-C, U as damage to the Test Set may result.

d. The insulation resistance in each case shall be 100 megohms minimum.

*4.4.5 Environmental sampling test. During the course of production, Test Sets shall be subject to the Vibration Test, Method 514, of MIL-STD-810. (See 4.3.2f.)

4.4.5.1 Sampling procedure. Sampling Procedures provided by MIL-STD-105 for special Inspection Level S-3, Single Sampling, Normal Inspection, Acceptable Quality Level (AQL) Value of 4.0 for Major Defects. An Operation Test (4.4.3) and Insulation Test (4.4.4) shall be performed before the start of and after completion of each of the above two tests. Quantitative data shall be recorded.

4.4.5.2 Inspection lot. An Inspection Lot shall consist of a maximum of one month's production but not less than 25 Test Sets.

4.4.5.3 Unit of product and sample selection. The Unit of Product for sampling purposes shall be one complete Test Set. Samples shall be selected in a random manner from the production lot. The lot identity shall be maintained during performance of this test, and no units from the lot shall be supplied prior to determination of acceptability of the sample.

4.4.5.4 Defect classification. Each unit which fails to meet one or more of the specified functional or mechanical requirements as a result of the Vibration Test shall be deemed to be one Major Defect. No defect resulting from this test shall be classified as Critical or Minor.

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5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Level A. Each Test Set shall be packaged in accordance with Method IIb of MIL-P-116.

5.1.2 Level C. Test Sets shall be packaged in accordance with the manufacturer's commercial practice.

5.2 Packing.

5.2.1 Level A. Test Sets packaged in accordance with 5.1.1 shall be packed in containers conforming to PPP-B-591 Fiberboard, Wood-Cleated, overseas type, PPP-B-601, Wood, Cleated-plywood, overseas type, PPP-B-621 Wood, Nailed and Lock-corner, Class 2, PPP-B-636 Fiberboard, Weather Resistant or PPP-B-640 Fiberboard, Corrugated Triple-wall Class 2. Boxes shall be sealed, closed and strapped for overseas shipment in accordance with the applicable box specification and appendix thereto. Wood or cleated boxes shall be lined with a sealed waterproof case liner conforming to MIL-L-10547 except that when the exterior container of Method IIb packages has strength and water resistance as specified in PPP-B-636, a case liner will not be required. The gross weight of wood, wood-cleated or triple wall corrugated boxes shall not exceed 200 pounds, fiberboard boxes shall not exceed the weight limitation of the applicable box specification.

5.2.2 Level B. Test Sets packaged in accordance with 5.1.1 shall be packed in containers conforming to PPP-B-591 Fiberboard, Wood-cleated, Domestic Type, PPP-B-601, Wood Cleated-plywood, Domestic Type, PPP-B-621, Wood, Nailed and Lock-corner, Class 1, PPP-B-636 Fiberboard Domestic or PPP-B-640 Fiberboard Corrugated triple-wall, Class 1. Boxes shall be sealed and closed in accordance with the requirements of the applicable box specification. The gross weight of wood, wood-cleated or triple-wall corrugated boxes shall not exceed 250 pounds, fiberboard boxes shall not exceed the weight limitation of the applicable box specification.

5.2.3 Level C. Test Sets shall be packed in containers in a manner to insure acceptance and safe delivery at destination. Containers shall comply with the consolidated freight classification rules or other regulations as applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, marking for shipment shall be in accordance with MIL-STD-129.

5.4 Cure dates. The contractor shall comply with the requirements of ANA Bulletin 438 as applicable.

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*6. NOTES

*6.1 Intended use. This Test Set is intended to perform accurate and safe preflight operational tests on launchers that launch AGM-45A Shrike Missiles.

*6.2 Ordering data. Procurement documents should specify:

*6.2.1 Procurement requirements

a. Title, number, and date of this specification.

b. Levels of packaging and packing required (see 5.1 and 5.2).

c. Serial numbers applicable to contract quantities of Test Sets.

d. Compliance with MIL-I-45208, Inspection System Requirements.

*6.2.2 Contract data requirements. Data specified in 4.1.1 and 4.1.2 will be delivered as identified on a numbered DD Form 1664 when specified on a DD Form 1423 incorporated into the contract.

*6.3 First article. When first article is required, the contract or other procurement document should reflect the policy, instructions and clauses set forth in the Armed Services Procurement Regulation, Section I, Part 19, pages 198.55 thru 198.60, dated 1 January 1969.

*6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

CUSTODIAN:

Air Force - 84

PREPARING ACTIVITY:

Air Force - 84

REVIEWER:

Air Force - 70

PROJECT NUMBER:

4935 - F006

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE. AF-WP-O-MAY 64 505

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